

HOMework 1

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Problem 1. If r is rational ($r \neq 0$) and x is irrational, prove that $r + x$ and rx is irrational.

Proof. Suppose $r + x$ is rational. Let α be some rational. We have $r + x = \alpha \implies x = \alpha - r$. This is a contradiction, since an irrational cannot equal a rational difference. Now, using α , assume that rx is rational. We have $rx = \alpha \implies x = \frac{\alpha}{r}$ (using the fact $r \neq 0$). This is contradictory because an irrational cannot equal a rational quotient. $\square$

Problem 2. Let E be a nonempty subset of an ordered set; suppose α is a lower bound of E and β is an upper bound of E . Prove that $\alpha \leq \beta$.

Proof. Let $x \in E$. We are given that α is a lower bound of E , and β is an upper bound for E . By definition, $\alpha \leq x$, and $x \leq \beta$. Thus $\alpha \leq x \leq \beta \implies \alpha \leq \beta$. $\square$

Problem 3. Let A be a nonempty set of real numbers which is bounded below. Let $-A$ be the set of all numbers $-x$, where $x \in A$. Prove that

$$\inf A = -\sup(-A)$$

Proof. We are given that A is non-empty and bounded below. Let $\ell = \{x \in \mathbb{R} : \forall a \in A, x \leq a\}$. By the completeness axiom, we may deduce that $\alpha = \sup(\ell)$. This implies that $\alpha \leq a \implies \alpha = \inf(A)$. Now using the inequality, multiplying A by -1 , our inequality becomes $-\alpha \geq -a$. We know $-\alpha$ is an upper bound for $-A$. Let u be any upper bound for $-A$. Thus, $-\alpha \leq u \implies \alpha \geq -u$. So $-u$ is a lower bound for A . But $\alpha = \inf(A)$, so $\alpha \geq -u \implies -\alpha \leq u$. So $-\alpha = \sup(-A)$. Therefore, $\inf(A) = -\sup(-A) = \alpha$. $\square$

Problem 4. Suppose S is an ordered set with the greatest lower bound property. Show S has the least upper bound property.

Proof. Let U be a non-empty subset of S that is bounded above. Define $K = \{s \in S : s \text{ is an upper bound for } U\}$, where $K \neq \emptyset$ since U is known to be bounded. Using the greatest lower bound property, $\beta = \inf(K) \iff \forall k \in K, \beta \leq k$. Assume that for $s \in U, s > \beta$. We know that $\forall k \in K, s \leq k$. But $\inf(K) = \beta$ and is the greatest lower bound for K . So $s \leq \beta$, which is contradictory to our supposition. Therefore $s \leq \beta \implies \beta = \sup(U)$ shows that S has the least upper bound property. $\square$